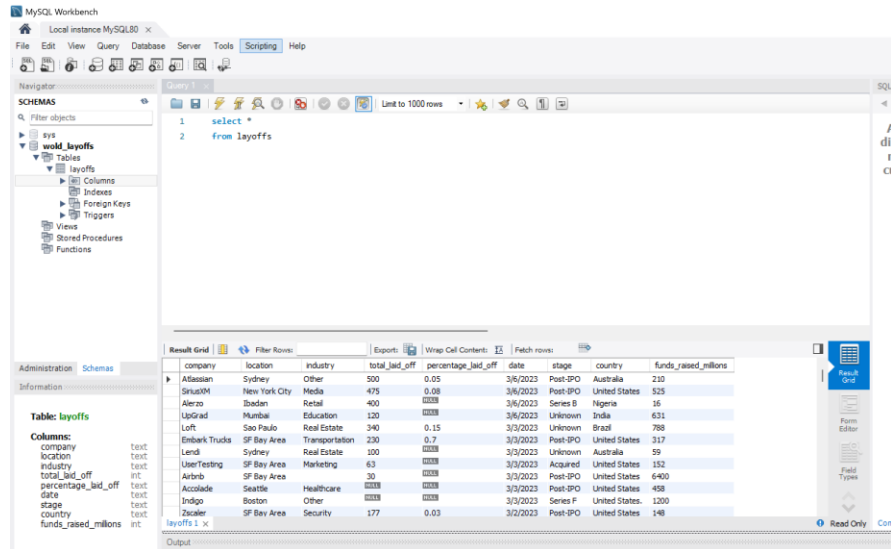


SQL Data Cleaning Project

SELECT*

FROM layoffs;



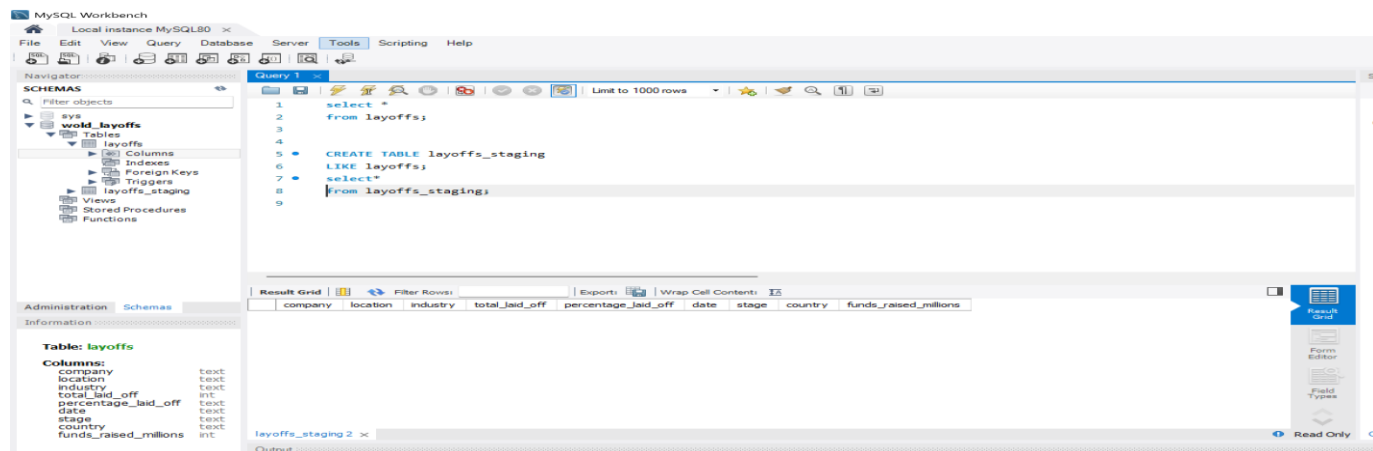
- Prior to making any changes, I create a backup copy of the original table.

CREATE TABLE layoffs_staging

LIKE layoffs;

SELECT *

FROM layoffs_staging;

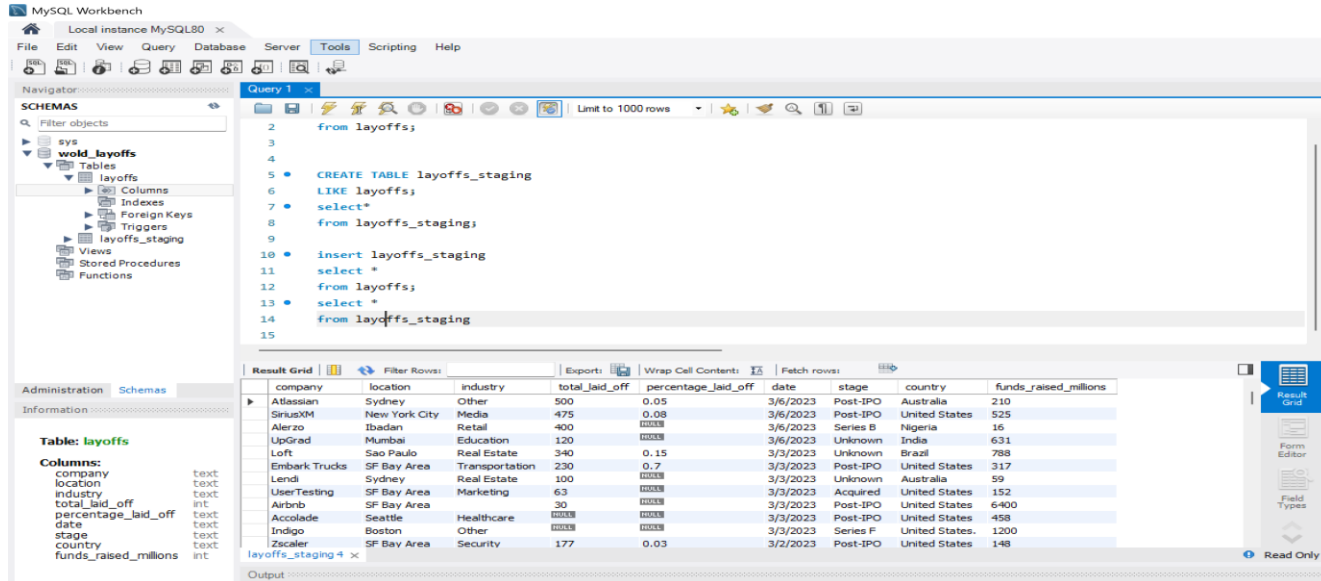


INSERT layoffs_staging SELECT *

FROM layoffs;

SELECT *

FROM layoffs_staging;



Detail:

- Using the same structure as layoffs, I make a new table called layoffs_staging.
- After that, I move all of the layoffs data into layoffs_staging.
- Lastly, I verify that the data has been accurately copied.

➤ Remove Duplicates

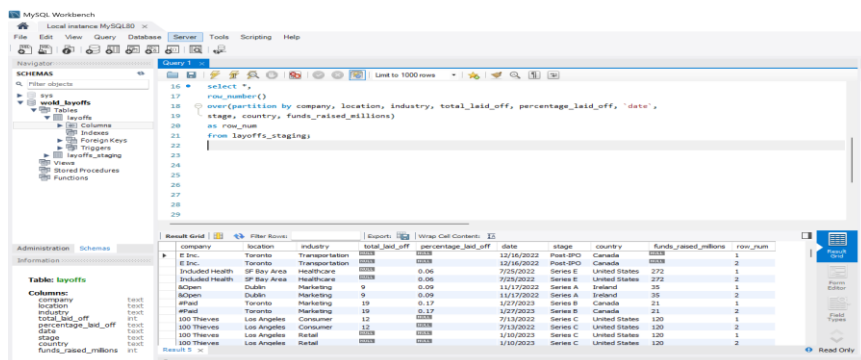
I use the ROW_NUMBER() function. SELECT *,

ROW_NUMBER()

OVER(PARTITION BY company, location, industry, total_laid_off, percentage_laid_off, `date`,

stage, country, funds_raised_millions) AS row_num

FROM layoffs_staging;



```

SELECT* , ROW_NUMBER()

OVER(PARTITION BY company, location, industry, total_laid_off, percentage_laid_off, `date` ,

stage, country, funds_raised_millions ) AS row_num FROM layoffs_staging)

SELECT*

FROM duplicate_cte WHERE row_num >1 ;

```

	company	location	industry	total_laid_off	percentage_laid_off	date	stage	country	funds_raised_millions	row_num
►	Casper	New York City	Retail	NULL	NULL	9/14/2021	Post-IPO	United States	339	2
	Cazoo	London	Transportation	750	0.15	6/7/2022	Post-IPO	United Kingdom	2000	2
	Hibob	Tel Aviv	HR	70	0.3	3/30/2020	Series A	Israel	45	2
	Wildlife Studios	Sao Paulo	Consumer	300	0.2	11/28/2022	Unknown	Brazil	260	2
	Yahoo	SF Bay Area	Consumer	1600	0.2	2/9/2023	Acquired	United States	6	2

- To get rid of duplicates, make a new table called layoffs_staging2 to hold the cleaned data.
- Make a copy of every record in layoffs_staging, giving duplicates a row number.
- Remove any additional duplicate records.

```

CREATE TABLE `layoffs_staging2` (
`company` text,
`location` text,
`industry` text,
`total_laid_off` int DEFAULT NULL,
`percentage_laid_off` text,
`date` text,
`stage` text,
`country` text,
`funds_raised_millions` int DEFAULT NULL,
`row_num` INT
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

```

```
SELECT *

FROM layoffs_staging2; INSERT INTO layoffs_staging2

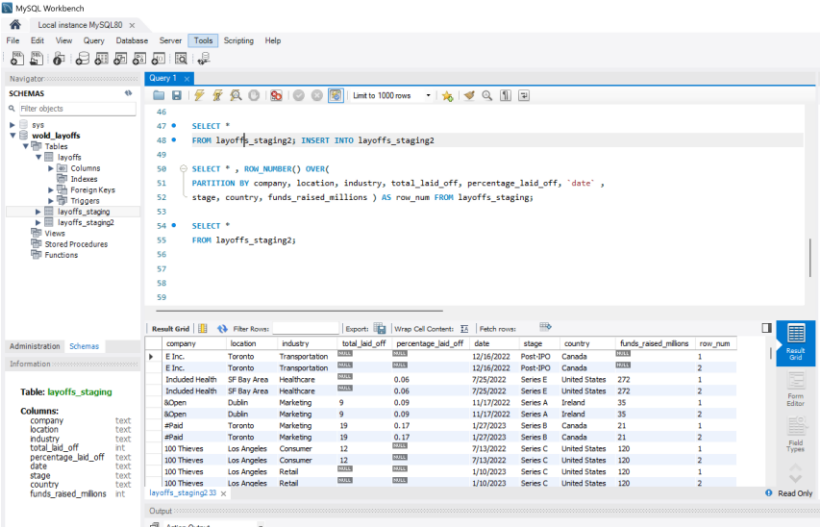
SELECT *, ROW_NUMBER() OVER(

PARTITION BY company, location, industry, total_laid_off, percentage_laid_off, `date` ,

stage, country, funds_raised_millions ) AS row_num FROM layoffs_staging;
```

```
SELECT *

FROM layoffs_staging2;
```

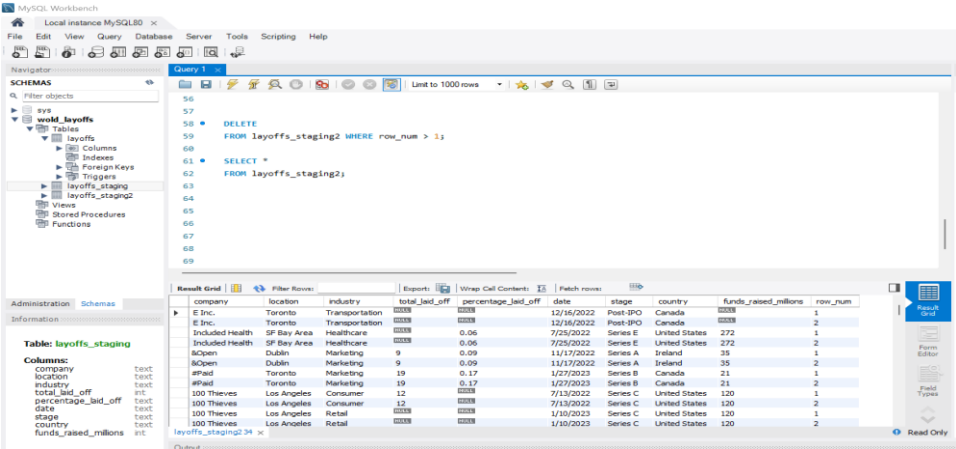


```
DELETE

FROM layoffs_staging2 WHERE row_num > 1;

SELECT *

FROM layoffs_staging2;
```



Detail:

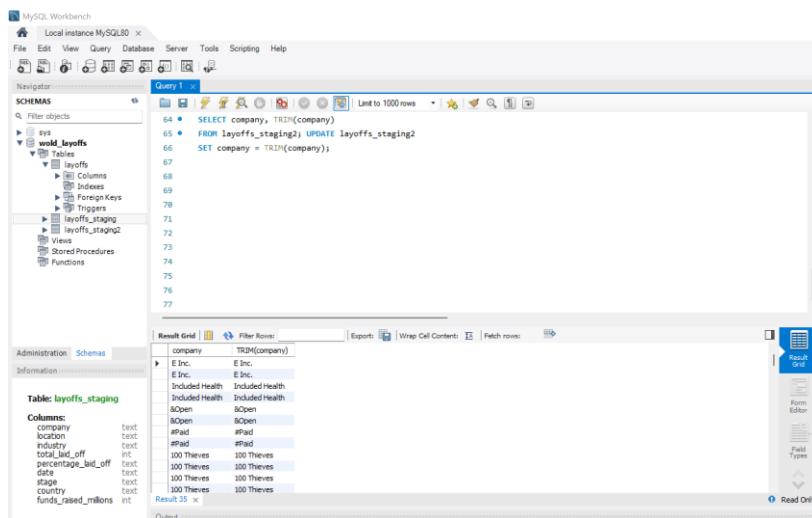
- I make layoffs_staging2, a second staging table.
- Duplicate rows are given a number using the ROW_NUMBER() method.
- I maintain only the first instance of each duplication and remove all rows where row_num > 1.

➤ Standardize Data

Remove white space from the company column and update it `SELECT company, TRIM(company)`

`FROM layoffs_staging2; UPDATE layoffs_staging2`

`SET company = TRIM(company);`

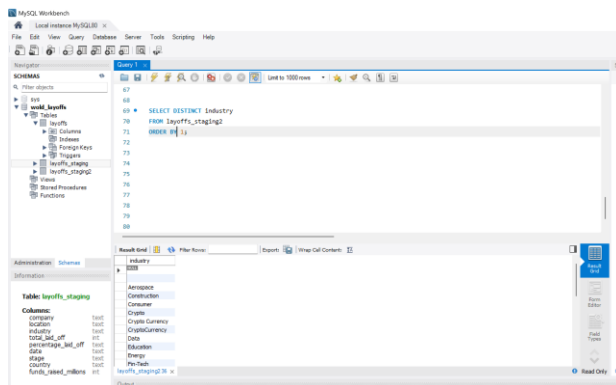


➤ Verify and fix any spelling mistakes in the industry table. CHOOSE A UNIQUE INDUSTRY

Select **DISTINCT** Industry

`FROM layoffs_staging2`

`ORDER BY 1;`



```

SELECT *

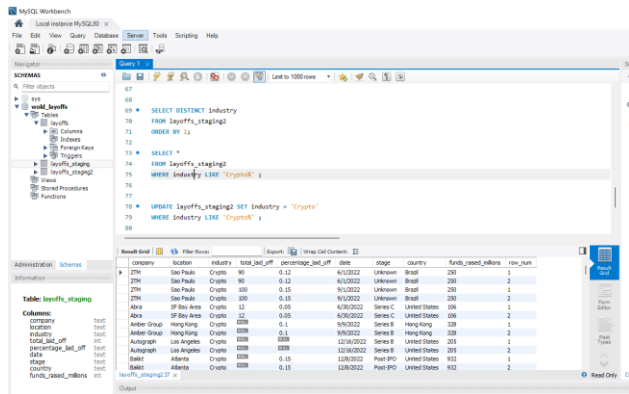
FROM layoffs_staging2

WHERE industry LIKE 'Crypto%' ;

UPDATE layoffs_staging2 SET industry = 'Crypto'

WHERE industry LIKE 'Crypto%' ;

```



- Analyzing for errors in the location table and correct where required.

```

SELECT DISTINCT location

FROM layoffs_staging2 ORDER BY 1;

UPDATE layoffs_staging2 SET country = 'United States'

WHERE country LIKE 'United States%' ;

```

Detail:

- TRIM(company): Removes extra spaces from the company names.
- LIKE 'Crypto%'

Step 4: Date Format

```

SELECT `date`,

STR_TO_DATE( `date` , '%m/%d/%Y' ) FROM layoffs_staging2 ;

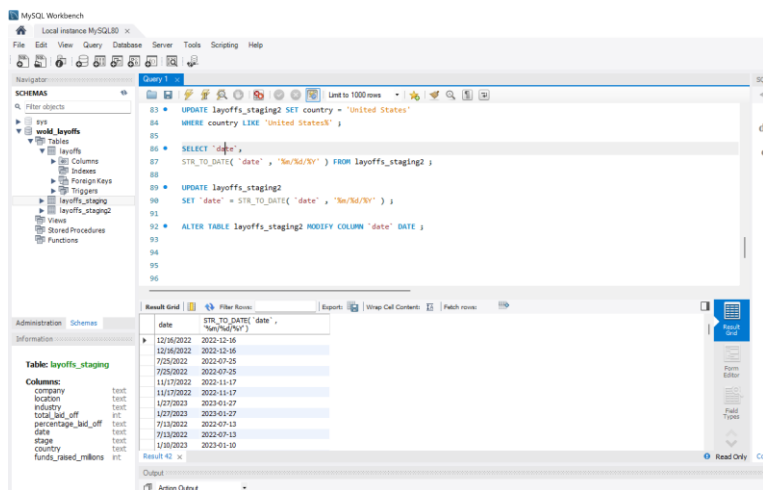
```

```

UPDATE layoffs_staging2

SET `date` = STR_TO_DATE( `date` , '%m/%d/%Y' ) ;

```

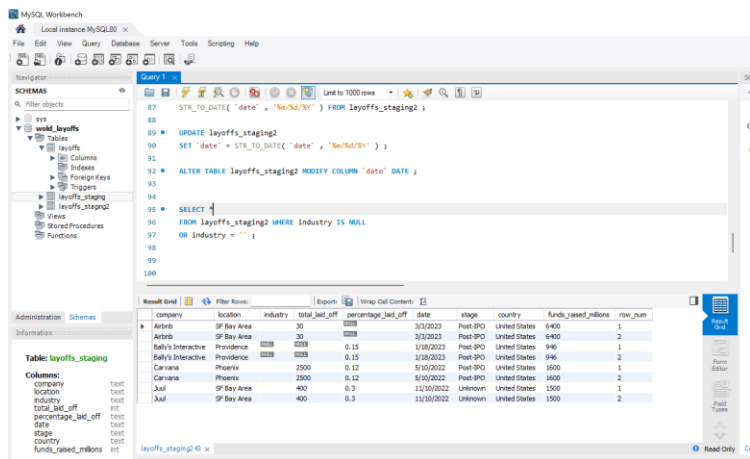


Detail:

1. STR_TO_DATE() converts text-based dates into proper date format.
2. ALTER TABLE changes the column type to DATE.

➤ Handle Missing Values

--Find missing values in industry column: SELECT *
FROM layoffs_staging2 WHERE industry IS NULL
OR industry = '' ;

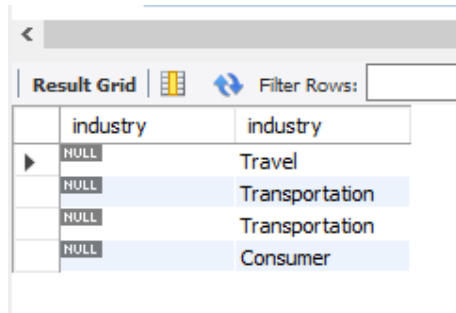


Replace blank values with NULL:

UPDATE layoffs_staging2

SET industry = null

WHERE industry = '' ;



	industry	industry
▶	NULL	Travel
	NULL	Transportation
	NULL	Transportation
	NULL	Consumer

- Fill missing industry names where possible:

SELECT *

FROM layoffs_staging2 WHERE company = 'Airbnb'

SELECT t1.industry, t2.industry FROM layoffs_staging2 AS t1

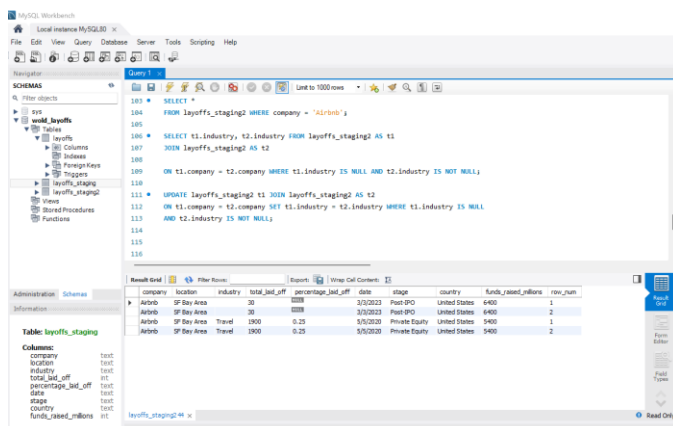
JOIN layoffs_staging2 AS t2

ON t1.company = t2.company WHERE t1.industry IS NULL AND t2.industry IS NOT NULL;

UPDATE layoffs_staging2 t1 JOIN layoffs_staging2 AS t2

ON t1.company = t2.company SET t1.industry = t2.industry WHERE t1.industry IS NULL

AND t2.industry IS NOT NULL;



```
101 SELECT *
102 FROM layoffs_staging2 WHERE company = 'Airbnb';
103
104 SELECT t1.industry, t2.industry FROM layoffs_staging2 AS t1
105 JOIN layoffs_staging2 AS t2
106 ON t1.company = t2.company WHERE t1.industry IS NULL AND t2.industry IS NOT NULL;
107
108 UPDATE layoffs_staging2 t1 JOIN layoffs_staging2 AS t2
109 ON t1.company = t2.company SET t1.industry = t2.industry WHERE t1.industry IS NULL
110 AND t2.industry IS NOT NULL;
```

company	location	industry	total_job_off	percentage_job_off	date	stage	country	funds_raised_millions	row_num
Airbnb	SF Bay Area	IT	1000	0.25	3/1/2013	Post IPO	United States	6400	1
Airbnb	SF Bay Area	IT	1000	0.25	3/1/2013	Post IPO	United States	6400	2
Airbnb	SF Bay Area	Travel	1000	0.25	5/1/2013	Private Equity	United States	9400	1
Airbnb	SF Bay Area	Travel	1000	0.25	5/1/2013	Private Equity	United States	9400	2

- Remove rows where key fields are missing:

SELECT *

FROM layoffs_staging2 WHERE total_laid_off IS NULL

AND percentage_laid_off IS NULL;

The screenshot shows the MySQL Workbench interface. The 'Query' tab is active, displaying the following SQL query:

```
SELECT *
FROM layoffs_staging2 WHERE total_laid_off IS NULL
AND percentage_laid_off IS NULL;
```

The 'Result Grid' shows the results of the query. The columns are: company, location, industry, total_laid_off, percentage_laid_off, date, stage, country, funds_raised_millions, and row_num. The results are as follows:

company	location	industry	total_laid_off	percentage_laid_off	date	stage	country	funds_raised_millions	row_num
E Inc.	Toronto	Transportation	0000	0000	12/16/2022	Post IPO	Canada	0000	1
E Inc.	Toronto	Transportation	0000	0000	12/16/2022	Post IPO	Canada	0000	2
100 Thieves	Los Angeles	Retail	0000	0000	1/10/2023	Series C	United States	120	1
100 Thieves	Los Angeles	Retail	0000	0000	1/10/2023	Series C	United States	120	2
Accolade	Seattle	Healthcare	0000	0000	3/2/2023	Post IPO	United States	458	1
Accolade	Seattle	Healthcare	0000	0000	3/2/2023	Post IPO	United States	458	2
Ada	Toronto	Support	0000	0000	2/1/2023	Series C	Canada	190	1
Ada	Toronto	Support	0000	0000	2/1/2023	Series C	Canada	190	2
Adara	San Francisco	Travel	0000	0000	3/1/2020	Series C	United States	67	1
Adara	San Francisco	Travel	0000	0000	3/1/2020	Series C	United States	67	2
Adia	Bogota	Finance	0000	0000	6/14/2022	Series C	Colombia	376	1
Adia	Bogota	Finance	0000	0000	6/14/2022	Series C	Colombia	376	2

DELETE

FROM layoffs_staging2 WHERE total_laid_off IS NULL

AND percentage_laid_off IS NULL ;

- Remove Columns which are not needed

ALTER TABLE layoffs_staging2

DROP COLUMN row_num ;

Final Result

SELECT *

FROM layoffs_staging2 ;

MySQL Workbench interface showing the final result of the query. The query window displays the following SQL:

```
SELECT *  
FROM layoffs_staging2 ;
```

The result grid shows the following data:

company	location	industry	total_laid_off	percentage_laid_off	date	stage	country	funds_raised_millions
E Inc.	Toronto	Transportation	1000	100%	12/16/2022	Post-IPO	Canada	1000
E Inc.	Toronto	Transportation	1000	100%	12/16/2022	Post-IPO	Canada	1000
Included Health	SF Bay Area	Healthcare	0.06	0.06	7/25/2022	Series E	United States	272
Included Health	SF Bay Area	Healthcare	0.06	0.06	7/25/2022	Series E	United States	272
Open	Dublin	Marketing	9	0.09	11/17/2022	Series A	Ireland	35
Open	Dublin	Marketing	9	0.09	11/17/2022	Series A	Ireland	35
paid	Toronto	Marketing	19	0.17	1/27/2023	Series B	Canada	21
paid	Toronto	Marketing	19	0.17	1/27/2023	Series B	Canada	21
100 Thieves	Los Angeles	Consumer	12	100%	7/13/2022	Series C	United States	120
100 Thieves	Los Angeles	Consumer	12	100%	7/13/2022	Series C	United States	120
100 Thieves	Los Angeles	Retail	1000	100%	1/10/2023	Series C	United States	120